SAIKAT MONDAL, Ph.D.

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RESEARCH VISION AND CAREER OBJECTIVES

Saikat's research vision, **Human-Aligned Code Intelligence (HACI)**, seeks to advance **trustworthy and explainable AI for Software Engineering (SE)**. He studies how developers reason about code and builds AI techniques for **software maintenance and evolution**, focusing on **human-aligned code generation**, **evidence-grounded reasoning**, and **reliable AI agents** that balance autonomy with human oversight. His goal is to deliver solutions that are **robust, usable, and sustainable**.

To realize this vision, he aims to (1) **lead an internationally recognized research program** and train future researchers in empirical SE, AI for SE, and responsible AI, (2) **translate research into practical tools** through industry and interdisciplinary collaborations, and (3) promote **equity, diversity, and inclusion** through equitable recruitment, accessible research practices, and **inclusive teaching and mentorship**.

EDUCATION

Doctor of Philosophy, Computer Science/Software Engineering
University of Saskatchewan, Canada
May 2020 – July 2026
Dissertation: AI-Augmented Quality Engineering for Developer Knowledge Artifacts in Human-AI Ecosystems
Advisor: Dr. Chanchal K. Roy
Distinction: **Governor General's Gold Medal Nominee** (sole departmental nominee, Fall 2026) + Course Average: **93.5%**
Selected Honours: **IEEE TCSE Distinguished Paper Award (SANER 2025, top 4 of 52 accepted papers)** + **Best Paper Award (ISEC 2022)** + **Dr. Keith Geddes Award 2022 (Ph.D. Student of the Year)** + **Carl McCrosky Innovation Scholarship 2023** + **GSA Research Excellence in STEM Award 2022** + **Citizenship Award 2021**

Master of Science, Computer Science/Software Engineering
University of Saskatchewan, Canada
May 2018 – April 2020
Thesis: Investigating the Quality Aspects of Crowd-Sourced Developer Forum: A Case Study of Stack Overflow
Advisor: Dr. Chanchal K. Roy
Distinction: Course Average **92.0%**
Selected Honours: **University of Saskatchewan Graduate Thesis Award 2020 (University-wide Best M.Sc. Thesis)** + **Research Excellence Award 2020 (Best M.Sc. Thesis, Dept. of CS)** + **Dr. Keith Geddes Award 2019 (M.Sc. Student of the Year)**

Bachelor of Science, Computer Science & Engineering
Khulna University, Bangladesh
January 2005 – August 2010
Thesis: A New Approach of Extendable Multicast Routing Protocol in Mobile Ad Hoc Networks (published at ICCIT 2010)
Advisor: Dr. Md Saidur Rahman
Distinction: **Ranked 1st in class** + Merit List Scholarship, 2005–2010

PUBLICATIONS (37)

[Under Revision and Under Review]

[R5] **Saikat Mondal**, Banani Roy, Foutse Khomh, and Chanchal K. Roy. Perception and Reality: Attribution, Originality, Plagiarism, and Ownership in AI-Generated Code. *49th IEEE/ACM International Conference on Software Engineering (**ICSE 2027**)*, pp. 12, Dublin, Ireland, April 25 – May 1, 2027 (*Under Major Revision*)

[R4] Suborno Deb Bappon, **Saikat Mondal**, Chanchal K. Roy, and Kevin Schneider. Human-Aligned Enhancement of Programming Answers with LLMs Guided by User Feedback. *ACM Transactions on Software Engineering and Methodology (**TOSEM**)* (Impact Factor = **6.30**), pp. 29 (*Under Major Revision*)

[R3] Priata Nowshin, **Saikat Mondal**, Debajyoti Mondal, and Chanchal K. Roy. Policy Development for LLM-Assisted Bug Fixing: Practitioners' Concerns, Disclosure, Oversight, and Organizational Equilibria. *ACM Transactions on Software Engineering and Methodology (**TOSEM**)* (Impact Factor = **6.30**), pp. 34 (*Under Major Revision*) 🏆 **Best Poster Award at ICSAC 2025**

[R2] Joy Krishan Das, **Saikat Mondal**, and Chanchal K. Roy. Developers' Engagement and Perceptions of ChatGPT in GitHub Issue Resolution: A Mixed-Methods Study. *Empirical Software Engineering Journal (**EMSE**)* (Impact Factor = **4.50**), pp. 34 (*Under Major Revision*)

[R1] **Saikat Mondal** and Chanchal K. Roy. Characterizing the Effects of Revisions on Stack Overflow Answers: A Multi-Dimensional Quality Analysis. *Empirical Software Engineering Journal (**EMSE**)* (Impact Factor = **4.50**), pp. 50 (*Under Review*)

[2026]

[37] Al Muttakin, **Saikat Mondal**, and Chanchal K. Roy. The State of Open Science in Software Engineering Research: A Case Study of ICSE Artifacts. In *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**)*, pp. 2299–2310, Rio de Janeiro, Brazil, April 2026 (Acceptance rate: **21.85%**) ★ ACM Artifact Badges by peer reviews: **Functional** + **Available** + **Reusable**

[36] Shahriar Rumi Dipto, **Saikat Mondal**, and Chanchal K. Roy. Algorithm-Based Pipeline for Reliable and Intent-Preserving Code Translation with LLMs. In *Proceedings of the 34th IEEE/ACM International Conference on Program Comprehension (**ICPC 2026**)*, pp. 390–401, Rio de Janeiro, Brazil, April 2026 (Acceptance rate: **28.50%**) 🏆 **SOAR Distinguished Research Award 2025**

[35] Khairul Alam, **Saikat Mondal**, and Banani Roy. Why Are AI Agent–Involved Pull Requests (Fix-Related) Remain Unmerged? An Empirical Study. In *Proceedings of the 23rd IEEE/ACM International Conference on Mining Software Repositories (**MSR 2026** – Mining Challenge)*, pp. 964–968, Rio de Janeiro, Brazil, April 2026

[34] Al Muttakin, **Saikat Mondal**, and Chanchal K. Roy. OpenART: Manuscript-Driven Support for Promoting Executability of Software Engineering Artifacts. *1st IEEE/ACM Journal Ahead Workshop (**JAWs 2026**)*, pp. 5, Rio de Janeiro, Brazil, April 2026 (*Presented*) (Invited for **TOSEM** submission)

[33] Khairul Alam, **Saikat Mondal**, and Banani Roy. SciFlowgent: Agentic AI Support for Developing Scientific Workflows. *1st IEEE/ACM Journal Ahead Workshop (**JAWs 2026**)*, pp. 10, Rio de Janeiro, Brazil, April 2026 (*Presented*) (Invited for **TOSEM** submission)

[32] **Saikat Mondal**, Gias Uddin, and Chanchal K. Roy. Automatic Assistance to Mitigate Rollback Inconsistencies in Collaborative Edits. *Automated Software Engineering Journal* (Impact Factor = **3.90**), vol. 33, no. 1, pp. 54, 2026.

[2025]

[31] **Saikat Mondal**, Chanchal K. Roy, Hong Wang, Juan Arguello, and Samantha Mathan. Can We Trust the AI Pair Programmer? Copilot for API Misuse Detection and Correction. In *Proceedings of the 35th IEEE International Conference on Collaborative Advances in Software and Computing (**CASCON 2025**)*, pp. 56–61, Toronto, Canada, November 2025

[30] **Saikat Mondal** and Chanchal K. Roy. Does Editing Improve Answer Quality on Stack Overflow? A Data-Driven Investigation. In *Proceedings of the 41st IEEE International Conference on Software Maintenance and Evolution (**ICSME 2025**)*, pp. 492–504, Auckland, New Zealand, September 2025 🏆 **Distinguished Research Award at ICSAC 2025** ★ Invited for **EMSE** special issue

[29] Saumendu Roy, **Saikat Mondal**, Banani Roy, and Chanchal K. Roy. From Questions to Insights: Exploring XAI Challenges Reported on Stack Overflow Questions. In *Proceedings of the 29th International Conference on Evaluation and Assessment in Software Engineering (**EASE 2025**)*, pp. 261–271, Istanbul, Turkey, June 2025 (Acceptance rate: **27.00%**) ★ USask EDI Initiative Workshop 2025 – **Honorable Mention**

[28] **Saikat Mondal** and Chanchal K. Roy. GENCNIPPET: Automated Generation of Code Snippets for Supporting Programming Questions. In *Proceedings of the 22nd IEEE/ACM International Conference on Mining Software Repositories (**MSR 2025** – Registered Reports Track)*, pp. 8, Ottawa, Canada, April 2025

[27] Shamse Tasnim Cynthia, **Saikat Mondal**, Joy Krishan Das, and Banani Roy. Gender Disparities in Contributions, Leadership, and Collaboration: An Exploratory Study on Software Systems Research. In *Proceedings of the Sixth IEEE/ACM Workshop on Gender Equality, Diversity, and Inclusion in Software Engineering (**GEICSE 2025**)*, pp. 25–32, Ottawa, Canada, April 2025

[26] Joy Krishan Das, **Saikat Mondal**, and Chanchal K. Roy. Why Do Developers Engage with ChatGPT in Issue-Tracker? Investigating Usage and Reliance on ChatGPT-Generated Code. In *Proceedings of the 32nd IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2025**)*, pp. 68–79, Montreal, Canada, March 2025 (Acceptance rate: **23.11%**) 🏆 **IEEE TCSE Distinguished Paper Award** 🏆 **Distinguished Research Award at ICSAC 2025** ★ Invited for **EMSE** special issue ★ **Featured** by the Ministry of AI

[2024]

[25] Suborno Deb Bappon, **Saikat Mondal**, and Banani Roy. AUTOGENICS: Automated Generation of Context-Aware Inline Comments for Code Snippets on Programming Q&A Sites Using LLM. In *Proceedings of the 24th IEEE International Conference on Source Code Analysis & Manipulation (**SCAM 2024**)*, pp. 24–35, Flagstaff, AZ, USA, October 2024

[24] **Saikat Mondal** and Banani Roy. Reproducibility of Issues Reported in Stack Overflow Questions: Challenges, Impact & Estimation. *Journal of Systems and Software (**JSS**)* (Impact Factor = **3.80**), vol. 217, pp. 1–17, 2024.

[23] M. Raihan, Sami Azam, Laboni Akter, Md. Mehedi Hassan, Ryana Quadir, Asif Karim, **Saikat Mondal**, and Arun More. Risk Factors Categorizations of Ischemic Heart Disease in South-Western Bangladesh. *Data Intelligence* (Impact Factor = **1.30**), vol. 6, no. 3, pp. 834–868, 2024.

[22] **Saikat Mondal**, Mohammad Masudur Rahman, and Chanchal K. Roy. Can We Identify Stack Overflow Questions Requiring Code Snippets? Investigating the Cause & Effect of Missing Code Snippets. In *Proceedings of the 31st IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2024**)*, pp. 764–775, Rovaniemi, Finland, March 2024 (Acceptance rate: **25.60%**) 🏆 **SOAR Distinguished Research Award 2024**

[21] **Saikat Mondal**, Suborno Deb Bappon, and Chanchal K. Roy. Enhancing User Interaction in ChatGPT: Characterizing and Consolidating Multiple Prompts for Issue Resolution. In *Proceedings of the 21st IEEE/ACM International Conference on Mining Software Repositories (**MSR 2024** – Mining Challenge)*, pp. 222–226, Lisbon, Portugal, April 2024

[20] Joy Krishan Das, **Saikat Mondal**, and Chanchal K. Roy. Investigating the Utility of ChatGPT in the Issue Tracking System: An Exploratory Study. In *Proceedings of the 21st IEEE/ACM International Conference on Mining Software Repositories (**MSR 2024** – Mining Challenge)*, pp. 217–221, Lisbon, Portugal, April 2024

[2023]

[19] **Saikat Mondal**, Debajyoti Mondal, and Chanchal K. Roy. Investigating Technology Usage Span by Analyzing Users' Q&A Traces in Stack Overflow. In *Proceedings of the 30th IEEE Asia-Pacific Software Engineering Conference (**APSEC 2023**)*, pp. 589–593, Seoul, Korea, December 2023

[18] **Saikat Mondal**, Mohammad Masudur Rahman, and Chanchal K. Roy. Do Subjectivity and Objectivity Always Agree? A Case Study with Stack Overflow Questions. In *Proceedings of the 20th IEEE/ACM International Conference on Mining Software Repositories (**MSR 2023**)*, pp. 389–401, Melbourne, Australia, May 2023 (Acceptance rate: **37.00%**)

[2022]

[17] Saumendu Roy, Gabriel Laberge, Banani Roy, Foutse Khomh, Amin Nikanjam, and **Saikat Mondal**. Why Don't XAI Techniques Agree? Characterizing the Disagreements Between Post-hoc Explana-

tions of Defect Predictions. In *Proceedings of the 38th IEEE International Conference on Software Maintenance and Evolution (**ICSME 2022** – NIER Track)*, pp. 444–448, Limassol, Cyprus, October 2022

[16] **Saikat Mondal**, Gias Uddin, and Chanchal K. Roy. Automatic Prediction of Rejected Edits in Stack Overflow. *Empirical Software Engineering Journal (**EMSE**)* (Impact Factor = **4.50**), vol. 28, no. 1, pp. 1–43, 2022 (**ICSE 2023** Journal First Track) 🏆 **Carl McCrosky Innovation Scholarship 2023** 🏆 **SOAR Distinguished Research Award 2025** ★ Selected as the **weekly top article** in the Artificial Intelligence category in Doctor Penguin

[15] **Saikat Mondal**, Mohammad Masudur Rahman, Chanchal K. Roy, and Kevin Schneider. The Reproducibility of Programming-Related Issues in Stack Overflow Questions. *Empirical Software Engineering Journal (**EMSE**)* (Impact Factor = **4.50**), vol. 27, no. 3, pp. 1–52, 2022.

[14] **Saikat Mondal** and Banani Roy. Reproducibility Challenges and Their Impacts on Technical Q&A Websites: The Practitioners' Perspectives. In *Proceedings of the 15th ACM Innovations in Software Engineering Conference (**ISEC 2022**)*, pp. 1–11, DA-IICT Gandhinagar, India, February 2022 (Acceptance rate: **25.71%**) 🏆 **Best Paper Award**

[2021]

[13] **Saikat Mondal**, Gias Uddin, and Chanchal K. Roy. Rollback Edit Inconsistencies in Developer Forum. In *Proceedings of the 18th IEEE/ACM International Conference on Mining Software Repositories (**MSR 2021**)*, pp. 380–391, Madrid, Spain, May 2021

[12] **Saikat Mondal**, CM Khaled Saifullah, Avijit Bhattacharjee, Mohammad Masudur Rahman, and Chanchal K. Roy. Early Detection and Guidelines to Improve Unanswered Questions on Stack Overflow. In *Proceedings of the 14th ACM Innovations in Software Engineering Conference (**ISEC 2021**)*, pp. 1–11, Bhubaneswar, India, February 2021 (Acceptance rate: **33.33%**)

[2020]

[11] **Saikat Mondal**, Gias Uddin, and Chanchal K. Roy. Automatic Identification of Rollback Edit with Reasons in Stack Overflow Q&A Site. In *Proceedings of the 36th IEEE International Conference on Software Maintenance and Evolution (**ICSME 2020** – Registered Reports Track)*, pp. 856, Adelaide, Australia, September 2020

[2019]

[10] **Saikat Mondal**, Mohammad Masudur Rahman, and Chanchal K. Roy. Can Issues Reported at Stack Overflow Questions be Reproduced? An Exploratory Study. In *Proceedings of the 16th IEEE/ACM International Conference on Mining Software Repositories (**MSR 2019**)*, pp. 479–489, Montreal, Canada, May 2019 (Acceptance rate: **25.40%**) 🏆 **CSBS People's Choice Award 2021**

[9] M. Raihan, Parichay Kumar Mandal, Muhammad Muinul Islam, Tanvir Hossain, Promila Ghosh, Shakil Ahmed Shaj, Abdullah Anik, Mubtasim Rafid Chowdhury, **Saikat Mondal**, and Arun More. Risk Prediction of Ischemic Heart Disease Using Artificial Neural Network. In *Proceedings of the 2019 International Conference on Electrical, Computer and Communication Engineering (ECCE 2019)*, pp. 1–5, Cox's Bazar, Bangladesh, February 2019

[2010–2018]

[8] M. Raihan, Muhammad Muinul Islam, Promila Ghosh, Shakil Ahmed Shaj, Mubtasim Rafid Chowdhury, **Saikat Mondal**, and Arun More. A Comprehensive Analysis on Risk Prediction of Acute Coronary Syndrome Using Machine Learning Approaches. In *Proceedings of the 21st International Conference of Computer and Information Technology (ICCIT 2018)*, pp. 1–6, Dhaka, Bangladesh, December 2018

[7] Procheta Nag, **Saikat Mondal**, Foysal Ahmed, Arun More, and M. Raihan. A Simple Acute Myocardial Infarction (Heart Attack) Prediction System Using Clinical Data and Data Mining Techniques. In *Proceedings of the 20th International Conference of Computer and Information Technology (ICCIT 2017)*, pp. 1–6, Dhaka, Bangladesh, December 2017

[6] Anupam Kumar Bairagi, **Saikat Mondal**, and Debashish Chakroborti. Securing Bangla Text Communication Using Image Steganography with Dynamic Substitution in IoT Environment. *Khulna University Studies (KUS) Journal*, vol. 14, no. 1&2, pp. 95–106, 2017.

[5] M. Raihan, **Saikat Mondal**, Arun More, Md Omar Faruqe Sagor, Gopal Sikder, Mahbub Arab Majumder, Mohammad Abdullah Al Manjur, and Kushal Ghosh. Smartphone Based Ischemic Heart

Disease (Heart Attack) Risk Prediction using Clinical Data and Data Mining Approaches, a Prototype Design. In *Proceedings of the 19th International Conference on Computer and Information Technology (ICCIT 2016)*, pp. 299–303, Dhaka, Bangladesh, December 2016

[4] **Saikat Mondal**, Rameswar Debnath, and Borun Kumar Mondal. An Improved Color Image Steganography Technique in Spatial Domain. In *Proceedings of the 9th International Conference on Electrical and Computer Engineering (ICECE 2016)*, pp. 582–585, Dhaka, Bangladesh, December 2016

[3] Anupam Kumar Bairagi, **Saikat Mondal**, and Rameswar Debnath. A Robust RGB Channel Based Image Steganography Technique using a Secret Key. In *Proceedings of the 16th International Conference on Computer and Information Technology (ICCIT 2013)*, pp. 81–87, Khulna, Bangladesh, March 2014

[2] Anupam Kumar Bairagi, **Saikat Mondal**, and Amit Kumar Mondal. A Dynamic Approach in Substitution Based Audio Steganography. In *Proceedings of the International Conference on Informatics, Electronics & Vision (ICIEV 2012)*, pp. 501–504, Dhaka, Bangladesh, May 2012

[1] Md Saidur Rahman, **Saikat Mondal**, Shushanto Kumar Ghosh, and Md Mahbubur Rahman. A New Approach of Extendable Multicast Routing Protocol in MANET. In *Proceedings of the 13th International Conference on Computer and Information Technology (ICCIT 2010)*, pp. 120–124, Dhaka, Bangladesh, December 2010

AWARDS AND HONOURS (32)

[32] **[2026] Governor General's Gold Medal Nomination:** Sole nominee from the Department of Computer Science, University of Saskatchewan, for the Fall 2026 Governor General's Gold Medal, in recognition of outstanding academic and research achievement.

[31] **[2025] IEEE TCSE Distinguished Paper Award (SANER 2025):** Awarded to the top 4 of 52 accepted papers at the IEEE International Conference on Software Analysis, Evolution and Reengineering for *"Why Do Developers Engage with ChatGPT in Issue-Tracker? Investigating Usage and Reliance on ChatGPT-Generated Code"*.

[30] **[2025] SOAR Distinguished Research Award 2025 (*Honorable Mention*):** Awarded to the top five research presentations at the SOAR Symposium 2025. Award value: $350.

[29] **[2025] Distinguished Research Award 2025 (*2nd Runner-up*):** Awarded at the 9th Symposium on Innovations in Computer Science and Applied Computing (ICSAC 2025). Award value: $300.

[28] **[2025] BSAUS Research Excellence Award 2025:** Awarded annually by the Bangladeshi Students' Association at the University of Saskatchewan (BSAUS) to one undergraduate or graduate student for outstanding research and academic achievement.

[27] **[2024] SOAR Distinguished Research Award 2024:** Awarded to the top six research presentations at the SOAR Symposium 2024, held with the 8th Symposium on Innovations in Computer Science and Applied Computing (ICSAC 2024). Award value: $300.

[26] **[2024] Carl McCrosky Innovation Scholarship 2023:** Awarded to *only one* student by the Department of Computer Science, University of Saskatchewan, for innovative problem-solving and outstanding academic performance. Award value: $1,500.

[25] **[2023] Dr. Keith Geddes Award 2022 – Ph.D. (*Student of the Year*):** Awarded to *only one* Ph.D. student by the Department of Computer Science, University of Saskatchewan, for outstanding research and academic performance. Award value: $2,500.

[24] **[2023] Graduate Travel Award:** Awarded by the University of Saskatchewan for travel to ICSE 2023 in Melbourne, Australia. Award value: $550.

[23] **[2022] Best Paper Award (ISEC 2022):** Awarded by the 15th Innovations in Software Engineering Conference for *"Reproducibility Challenges and Their Impacts on Technical Q&A Websites: The Practitioners' Perspectives"*.

[22] **[2022] GSA Research Excellence in STEM Award 2022:** Awarded to *only one* student at the University of Saskatchewan for outstanding contributions to STEM research, research collaboration, and leadership.

[21] **[2022] Citizenship Award 2021:** Awarded by the Department of Computer Science, University of Saskatchewan, for academic and leadership excellence, organizing campus events, and volunteer service to the department.

[20] **[2022] Dr. Keith Geddes Award Nomination 2021 – Ph.D.:** Nominated by the Department of Computer Science, University of Saskatchewan, for outstanding research and academic performance in the Ph.D. program.

[19] **[2022] Carl McCrosky Innovation Scholarship Nomination 2021:** Nominated by the Department of Computer Science, University of Saskatchewan, for innovative research and academic performance in the Ph.D. program.

[18] **[2021] University of Saskatchewan Graduate Thesis Award 2020 (*University-wide Best M.Sc. Thesis*):** Awarded to *only one* M.Sc. student for the best thesis in the Physical and Engineering Sciences. Award value: $500.

[17] **[2021] Research Excellence Award 2020 (*Best M.Sc. Thesis*):** Awarded to *only one* M.Sc. student by the Department of Computer Science, University of Saskatchewan, for outstanding research and academic performance.

[16] **[2021] People's Choice Award:** Awarded at the Computer Science and Bioinformatics Symposium, University of Saskatchewan, for the poster *"The Reproducibility of Programming-Related Issues in Stack Overflow Questions"*.

[15] **[2021] Vanier Canada Graduate Scholarship Nomination:** Nominated by the Department of Computer Science, University of Saskatchewan, for research excellence, academic performance, and leadership.

[14] **[2020] Dr. Keith Geddes Award 2019 – M.Sc. (*Student of the Year*):** Awarded to *only one* M.Sc. student by the Department of Computer Science, University of Saskatchewan, for outstanding research and academic performance. Award value: $2,500.

[13] **[2020] Dean's Scholarship Nomination:** Nominated by the Department of Computer Science, University of Saskatchewan, for outstanding research and academic performance.

[12] **[2020] Graduate Travel Award:** Awarded by the Bangladesh-Sweden Trust Fund to support travel from Bangladesh to Canada for graduate studies.

[11] **[2019] Graduate Travel Award:** Awarded by the University of Saskatchewan for travel to MSR 2019 in Montreal, Canada. Award value: $350.

[10] **[2019] Dr. Keith Geddes Award Nomination 2018 – M.Sc.:** Nominated by the Department of Computer Science, University of Saskatchewan, for outstanding research and academic performance in the M.Sc. program.

[9] **[2015] Best Faculty Award:** Awarded by the Computer Science and Engineering Discipline, Khulna University, among 12+ faculty members, for outstanding contributions to teaching and academic achievement.

[8] **[2014] Outstanding Service Award:** Awarded by the 16th International Conference on Computer and Information Technology (ICCIT 2013) for organizing the conference.

[7] **[2012] Exceptional Engineer Award:** Annual award among 400+ employees at Samsung R&D Institute Bangladesh for community contribution, including teaching computer skills to eleven street children in Dhaka as a volunteer with CAFFE.

[6] **[2012] Icon of the Month Award (*Best Employee*):** Awarded among 50+ employees at Samsung R&D Institute Bangladesh for outstanding contributions to research and development projects.

[5] **[2008] Programming Contest Award:** Awarded by Khulna University, Bangladesh, for placing *first runner-up* in the programming contest.

[4] **[2006] Math Olympiad Winner Award:** Awarded by Khulna University, Bangladesh, for winning the Math Olympiad as *champion*.

[3] **[2001] Best Student Award:** Awarded by the Member of Parliament, Khulna-1, Bangladesh, for obtaining the highest marks among about 5,000 students who took the Secondary School Certificate (SSC) exam from Khulna-1 in 2000. Award value: 10,000 BDT.

[2] **[2000] Champion and Best Speaker Awards:** Debate team champion and individual Best Speaker in the inter-school debate competition organized by Rupantar, Batiaghata, Khulna, Bangladesh.

[1] **[1990–2002] Cultural Competition Awards:** Received multiple awards for placing first in school- and college-level debate, poetry recitation, public speaking, and acting competitions.

RESEARCH GRANTS (8)

[8] **[2026] Lakehead University Start-up Fund**
Role Principal Investigator
Duration August 2026 – December 2028
Amount $30,000 ($10,000 research funds + $20,000 to support MSc students over two years)

[7] **[2026] NSERC Canada Postdoctoral Research Award (CPRA) (*Declined*)**
Title Human-Aligned Code Intelligence for Trustworthy and Explainable Software Solutions
Role Principal Applicant
Duration August 2026 – July 2028
Amount $140,000 ($70,000 $\times$ 2 years)

[6] **[2026] IVADO Postdoctoral Research Funding (*Declined*)**
Title Human-Aligned Code Intelligence for Trustworthy and Explainable Software Solutions
Role Principal Applicant
Duration August 2026 – July 2028
Amount $200,000 ($100,000 $\times$ 2 years)

[5] **[2026] NSERC Discovery Grant**
Title Human-Aligned Code Intelligence: Toward Trustworthy, Explainable, and Sustainable Software Engineering
Role Principal Applicant
Duration April 2027 – March 2032
Status Under review
Amount $448,000

[4] **[2026] New Frontiers in Research Fund (NFRF) – Exploration**
Title Transforming Housing Construction into a Programmable Production System
Role Co-investigator
Applicants Tharindu Dodanwala (PI)
Duration April 2027 – March 2029
Status Under review
Amount $243,750

[3] **[2026] New Frontiers in Research Fund (NFRF) – Exploration**
Title My Best Friend, AI? Exploring the Role of Conversational AI in Youth
Role Collaborator
Applicants Ronnie de Souza Santos (PI), Brae Anne McArthur, Kelly Schwartz, and Nadia Delanoy
Duration April 2027 – March 2029
Status Under review
Amount $250,000

[2] **[2026] Social Sciences and Humanities Research Council (SSHRC) Grant**
Title Can You Keep a Secret? Social and Relational Harms Among Youth in the Age of Generative AI
Role Collaborator
Applicants Ronnie de Souza Santos (PI) and Tonya Callaghan
Duration April 2027 – March 2031
Status Under review
Amount $219,000

[1] **[2017] Khulna University Research Grant**
Title Protein-Fold Recognition using Machine Learning Techniques from Amino Acid Sequence
Role Co-investigator
Applicants Dr. Rameswar Debnath (PI)
Duration May 2017 – April 2018
Amount $\approx$ $1,000

SCHOLARSHIPS (15)

[15] **[2025–2026] Faculty Scholarship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the Ph.D. program. Amount: $30,000.

[14] **[2024–2025] Faculty Scholarship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the Ph.D. program. Amount: $30,000.

[13] **[2023–2024] Faculty Scholarship and Graduate Teaching Fellowship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the Ph.D. program. Amount: $24,500.

[12] **[2022–2023] Faculty Scholarship and Graduate Teaching Fellowship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the Ph.D. program. Amount: $24,500.

[11] **[2022–2023] Faculty Scholarship (Top-up):** Awarded by the Department of Computer Science, University of Saskatchewan, for research excellence in the Ph.D. program. Amount: ≈ $3,000.

[10] **[2021–2022] Faculty Scholarship and Graduate Teaching Fellowship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the Ph.D. program. Amount: $24,500.

[9] **[2021] Faculty Scholarship (Top-up):** Awarded by the Department of Computer Science, University of Saskatchewan, for research excellence in the Ph.D. program. Amount: ≈ $2,500.

[8] **[2021] GSA Bursary:** Awarded by the Graduate Students' Association, University of Saskatchewan, for community involvement and strong academic performance. Amount: $750.

[7] **[2020–2021] Faculty Scholarship and Graduate Teaching Fellowship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the Ph.D. program. Amount: $24,500.

[6] **[2020] Faculty Scholarship and Graduate Teaching Fellowship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the Ph.D. program. Amount: $1,333 per month for four months.

[5] **[2019–2020] Faculty Scholarship (Top-up):** Awarded by the Department of Computer Science, University of Saskatchewan, for research excellence in the M.Sc. program. Amount: ≈ $8,000.

[4] **[2019] GSA Bursary:** Awarded by the Graduate Students' Association, University of Saskatchewan, for community involvement and strong academic performance. Amount: $1,000.

[3] **[2018–2020] Faculty Scholarship and Graduate Teaching Fellowship:** Awarded by the Department of Computer Science, University of Saskatchewan, for the M.Sc. program. Amount: $40,000 ($20,000 per year for two years).

[2] **[2005–2010] Merit List Scholarship:** Awarded by Khulna University, Bangladesh, for academic excellence in the B.Sc. program.

[1] **[2000–2002] Merit List Scholarship:** Awarded by the Government Education Board (Jessore), Bangladesh, for excellence in the 2000 Secondary School Certificate (SSC) exam.

EMPLOYMENT HISTORY

Assistant Professor, Department of Computer Science
August 2026–Present Lakehead University, Thunder Bay, Ontario, Canada

Responsibilities: Appointed as an Assistant Professor (limited term, three years) in the Department of Computer Science. Saikat's job responsibilities include (a) establishing an independent research program by securing research grants and supervising graduate and undergraduate students, (b) teaching undergraduate- and graduate-level courses, and (c) delivering administrative service.

Research Intern
March 2025–September 2025 NeuroAI, Canada

Responsibilities: Contributed to the BRIDGES Engage program, an AI-driven tool that provides automated, context-aware support to parents of children with special needs. BRIDGES Engage is the family-facing component of NeuroAI's broader solution, offering personalized intervention strategies, educational resources, and real-time Q&A through its AI assistant. Saikat built conversational flows and embedded large language models (LLMs) that interpret parent queries, respond with empathetic, evidence-based guidance, and surface resources aligned with each child's developmental profile.

Research Advisor
February 2023–December 2024 DreamOnline Limited, Bangladesh

Responsibilities: Worked with the company's AI team and guided them in developing automated tools and techniques that help developers produce high-quality software.

Graduate Research and Teaching Assistant, Department of Computer Science
September 2018–July 2026 University of Saskatchewan, Canada

Responsibilities: Saikat was responsible for (a) conducting research and development, (b) teaching undergraduate and graduate courses, (c) mentoring students and evaluating student projects, and (d) contributing to curriculum design, assignment development, and grading for undergraduate courses.

Research Assistant
August 2015–May 2018 Rural Health Progress Trust, Maharashtra, India

Responsibilities: Saikat was responsible for (a) conducting research on detecting heart diseases by analyzing clinical data, (b) collecting and processing data from online applications and heart camps, and (c) supervising the research team.

Assistant Professor, Department of Computer Science and Engineering
July 2015–April 2026 Khulna University, Bangladesh

Responsibilities: Appointed as a lecturer and later promoted to Assistant Professor. On approved study leave from May 2018 to April 2026 to pursue M.Sc. and Ph.D. studies at the University of Saskatchewan. Saikat was responsible for (a) teaching undergraduate classes, curricular innovation, course planning and evaluation, conducting exams, and publishing grades, (b) conducting research and co-supervising undergraduate theses, (c) conducting academic projects, organizing student contests and project festivals, and leading students in regional and national contests, (d) managing the department's computer club, library, and website, and (e) conducting administrative affairs such as admission test management and departmental purchase inspection.

Assistant Director of Students' Affairs
December 2012–April 2016 Khulna University, Bangladesh

Responsibilities: Appointed in addition to the faculty position. Saikat was responsible for (a) university students' welfare, (b) organizing cultural events and festivals, (c) celebrating national days, (d) advising cultural organizations of the university, and (e) promoting law and order at the university.

Lecturer, Department of Computer Science and Engineering
May 2012–July 2015 Khulna University, Bangladesh

Responsibilities: Appointed as a full-time faculty member. Saikat was responsible for (a) teaching undergraduate classes, curricular innovation, course planning and evaluation, conducting exams, and publishing grades, (b) conducting research, (c) conducting academic projects, organizing student contests and project festivals, and leading students in regional and national contests, (d) managing the department's computer club, library, and website, and (e) conducting administrative affairs such as admission test management and departmental purchase inspection. Received the **Best Faculty Award** among 12+ faculty members of the department.

Adjunct Faculty, Department of Computer Science and Engineering
May 2012–May 2018 North Western University, Bangladesh

Responsibilities: Appointed as a part-time faculty member. Saikat was responsible for (a) teaching undergraduate and diploma engineering classes, course evaluation, conducting exams, and publishing grades, and (b) supervising academic projects.

Software Engineer
November 2010–May 2012 Advanced Research and Development (now Solution Lab), Samsung R&D Institute Bangladesh Ltd.

Responsibilities: Selected to work in the advanced R&D division for strong research and development skills. Saikat was responsible for (a) solving research problems, developing software applications and systems, and reporting project updates, and (b) presenting research ideas on wireless connectivity and writing research proposals. Received the **Icon of the Month Award** among 50+ employees and the annual **Exceptional Engineer Award** among 400+ employees.

TEACHING EXPERIENCE

[Lakehead University]

[18] Fall 2026/**COMP 3415 – Software Engineering** (Instructor, Thunder Bay): *Enrollment* 41

[17] Fall 2026/**COMP 3415 – Software Engineering** (Instructor, Barrie STEM Hub): *Enrollment* 34

[16] Fall 2026/**COMP 4310 – Web Health Informatics** (Instructor): *Enrollment* 21

[University of Saskatchewan]

[15] Winter 2025/**CMPT 470/816 – Advanced Software Engineering** (Co-Instructor and Lead Coordinator): *Enrollment* 74

[14] Winter 2022/**CMPT 370 – Intermediate Software Engineering** (Project Mentor and Evaluator): 16 students

[Khulna University]

[13] **CSE 4221 – Pattern Recognition** (Instructor)

[12] **CSE 4205/4206 – Artificial Intelligence and Laboratory** (Instructor)

[11] **CSE 4105/4106 – Compiler Design and Laboratory** (Instructor)

[10] **CSE 3200 – Web Programming Project** (Instructor)

[9] **CSE 2201 – Algorithms** (Instructor)

[8] **CSE 2200 – Software Development Project** (Instructor)

[7] **CSE 2151 – Data Structures and Algorithms** (Instructor)

[6] **CSE 1201 – Object-Oriented Programming** (Instructor)

[5] **CSE 1151 – Computer Programming** (Instructor)

[4] **CSE 1103 – Structured Programming** (Instructor)

[*Syllabus of these courses can be found at the **website of Khulna University***]

[North Western University]

[3] **CSE 4301/4302 – Computer Graphics and Pattern Recognition and Laboratory** (Adjunct Instructor)

[2] **CSE 3207/3208 – Compiler Design and Laboratory** (Adjunct Instructor)

[1] **Software Development Project (Sessional)** (Adjunct Instructor)

[*Syllabus of these courses can be found at the **North Western University website***]

COURSE CURRICULUM DESIGN (3)

[3] **COMP 3415: Software Engineering** (Lakehead University, Fall 2026)

Overview A third-year course that combines foundational software engineering with contemporary AI-assisted and agentic practice. Students learn to plan, design, build, test, maintain, and evolve software through Agile processes, test-driven development and continuous integration, clean code and refactoring, UML and design patterns, and modern architectures including microservices and cloud-native systems. Dedicated modules cover AI-assisted engineering across the software lifecycle, agentic workflows with human oversight, and professional responsibility, including attribution, accountability, and the responsible use of AI. An Agile-based team project, organized in iterative sprints with user stories, testing, and continuous integration, takes students from requirements to a working system that evolves in response to change.

Role Redesign

Syllabus Course syllabus: Thunder Bay, Barrie STEM Hub

[2] **COMP 4310: Web Health Informatics** (Lakehead University, Fall 2026)

Overview A fourth-year course on how computing and information technologies support healthcare in Canada. Topics include health information systems, system analysis and design, usability and user-centered design, hospital information systems, electronic health records and Canadian digital health initiatives, virtual care, consumer and public health informatics, health analytics and AI, and the privacy and security of health information. A milestone-based project takes students from problem definition to a working prototype.

Role Redesign

Syllabus Course syllabus

[1] **CMPT 470/816: Advanced Software Engineering** (University of Saskatchewan, Winter 2025)

Overview A combined senior undergraduate and graduate course at the intersection of advanced software engineering and AI. Topics include source transformation with TXL, mining software repositories, AI for software engineering and software engineering for AI, prompt engineering, trustworthy and responsible AI, modern practices such as low-code/no-code and cloud-native development, software change management, process automation, quality assurance with automated testing and program repair, and design patterns. Assessment is built around a milestone-based group project.

Role Redesign

Syllabus Course syllabus

COLLABORATIONS (9)

[9] **Lakehead University, Canada**

Duration 2026 – Current

Collaborator Dr. Tharindu Dodanwala

Outcome New Frontiers in Research Fund (NFRF) – Exploration grant application as Co-investigator (under review)

[8] **University of Calgary, Canada**

Duration 2026 – Current

Collaborator Dr. Ronnie de Souza Santos

Outcome New Frontiers in Research Fund (NFRF) – Exploration and SSHRC grant applications as a collaborator (under review)

[7] **Singapore Management University, Singapore**

Duration 2024 – Current

Collaborator Dr. David Lo

Outcome Two ongoing projects: LLM-based detection and repair of security vulnerabilities (CWEs) in crowd-shared code and agent-based automatic program repair

[6] **University of Saskatchewan, Canada**

Duration 2023 – Current

Collaborator Dr. Debajyoti Mondal

Outcome APSEC x 1 and TOSEM x 1 (under revision)

[5] **University of Saskatchewan, Canada**

Duration 2022 – Current

Collaborator Dr. Banani Roy

Outcome ICSME x 1 (NIER), ISEC x 1, SCAM x 1, EASE x 1, GEICSE x 1, MSR x 1 (Mining Challenge), JAWs x 1, JSS x 1, and ICSE x 1 (under revision)

[4] **Polytechnique Montréal, Canada**

Duration 2022 – Current

Collaborator Dr. Foutse Khomh

Outcome ICSME x 1 (NIER) and ICSE x 1 (under revision)

[3] **York University, Canada**
Duration 2019 – Current
Collaborator Dr. Gias Uddin
Outcome MSR x 1, ICSME x 1 (Registered Report), EMSE x 1, and ASE Journal x 1

[2] **Dalhousie University, Canada**
Duration 2018 – Current
Collaborator Dr. Masud Rahman
Outcome MSR x 2, SANER x 1, ISEC x 1, and EMSE x 1

[1] **Rural Health Progress Trust, India**
Duration 2015 – 2018
Collaborator Dr. Arun More
Outcome ICCIT x 3, ECCE x 1, and Data Intelligence x 1

HQP TRAINING AND MENTORSHIP (13)

Mentored 11 graduate students (five Ph.D. and six M.Sc.) across the Software Research, Interactive Software Engineering, and Visualization, Geometry and Algorithms labs at the University of Saskatchewan, and co-supervised undergraduate research at the University of Saskatchewan and Khulna University.

[13] **Aditi Dhali (Ph.D.)**
Affiliation University of Saskatchewan, Canada
Duration 2026 – Current
Publications CogniCode: predicting human-perceived code quality using static analysis and machine learning (in preparation)

[12] **Shahriar Rumi Dipto (M.Sc.)**
Affiliation University of Saskatchewan, Canada
Duration 2025 – 2026
Publications ICPC 2026 x 1
Awards SOAR Distinguished Research Award 2025 (Champion)
Employment Siemens EDA, Saskatoon

[11] **Khairul Alam (Ph.D.)**
Affiliation University of Saskatchewan, Canada
Duration 2025 – Current
Publications MSR 2026 (Mining Challenge) x 1 and JAWs 2026 x 1

[10] **Al Muttakin (M.Sc.)**
Affiliation University of Saskatchewan, Canada
Duration 2024 – Current
Publications ICSE 2026 x 1, JAWs 2026 x 1, and TSE x 1 (in preparation)
Awards ACM Artifact Badges (Functional, Available, and Reusable), ICSE 2026

[9] **Priata Nowshin (M.Sc.)**
Affiliation University of Saskatchewan, Canada
Duration 2024 – Current
Publications TOSEM x 1 (under revision)
Awards Best Poster Award (ICSAC 2025) and Runner-up, EDI Initiative Workshop 2025, University of Saskatchewan
Employment Siemens EDA, Saskatoon

[8] **Joy Krishan Das (M.Sc.)**
Affiliation University of Saskatchewan, Canada
Duration 2023 – Current
Publications SANER 2025 x 1, MSR 2024 (Mining Challenge) x 1, and EMSE x 1 (under revision)
Awards IEEE TCSE Distinguished Paper Award (SANER 2025), Distinguished Research Award (Champion, ICSAC 2025), and SOAR Distinguished Research Award 2024
Employment Ph.D. Student, University of Waterloo

[7] **Suborno Deb Bappon (M.Sc.)**

Affiliation	University of Saskatchewan, Canada
Duration	2023 – Current
Publications	SCAM 2024 x 1, MSR 2024 (Mining Challenge) x 1, TOSEM x 1 (under revision)
Awards	Best Course Project (AUTOGENICS)
Employment	Siemens EDA, Saskatoon

[6] **Shamse Tasnim Cynthia (Ph.D.)**

Affiliation	University of Saskatchewan, Canada
Duration	2023 – Current
Publications	GEICSE 2025 x 1 and a journal article (in preparation)
Awards	SOAR Distinguished Research Award 2024, SOAR Woman of the Year Research Award 2024, and Winner, ICSAC 2024

[5] **Ajmain Inqiad Alam (Ph.D.)**

Affiliation	University of Saskatchewan, Canada
Duration	2023 – Current
Publications	JSS x 1 (in preparation)
Awards	CSAC Research Fest Award 2024 and SOAR Distinguished Research Award 2024

[4] **Saumendu Roy (Ph.D.)**

Affiliation	University of Saskatchewan, Canada
Duration	2022 – 2025
Publications	ICSME 2022 (NIER) x 1 and EASE 2025 x 1
Awards	SOAR Distinguished Research Award 2022 and Honorable Mention, EDI Initiative Workshop 2025, University of Saskatchewan
Employment	Postdoctoral Fellow, Concordia University

[3] **Sadman Amin (M.Sc.)**

Affiliation	University of Saskatchewan, Canada
Duration	2022 – Current
Publications	Code version migration challenges through the lens of neural analysis (in preparation)
Employment	Siemens EDA, Saskatoon

[2] **CMPT 470/816 Research Groups (18 students)**

Affiliation	University of Saskatchewan, Canada
Role	Research Project Mentor (Co-Instructor)
Duration	Winter 2025
Projects	Six research groups, guided through idea conceptualization, methodology, experiment design, data analysis, and report review
Publications	CASCON 2025 x 1

[1] **Undergraduate Thesis Students (13 students)**

Affiliation	Khulna University, Bangladesh
Role	Thesis Co-supervisor
Duration	2015 – 2018
Students	M. Raihan, Md Omar Faruqe Sagor, Tasmim Riza, Rufaida Kabir Zarin, Palash Sarker, Sk Arman, Mahadi Hasan Rakib, Mahdi Hussain, Mohoshin Ara Tahera, Anika Das, Farjana Yeasmin, Saumitra Bagchi, and Shahirul Kabir Sami
Topics	Information security (image and audio steganography) and heart disease prediction using machine learning
Publications	ICCIT x 3, ECCE x 1, and Data Intelligence x 1

PROGRAM COMMITTEE AND REVIEWING SERVICE

Program Committee Member (11)

[2027]

[11] **PC Member**: International Conference on Mining Software Repositories (MSR 2027) (Research Track)

[10] **PC Member**: International Conference on Mining Software Repositories (MSR 2027) (Mining Challenge Track)

[9] **PC Member**: International Conference on Software Analysis, Evolution and Reengineering (SANER 2027) (ERA Track)

[8] **Shadow PC Member**: International Conference on Software Engineering (ICSE 2027)

[2026]

[7] **PC Member**: International Conference on Software Analysis, Evolution and Reengineering (SANER 2026) (ERA Track)

[6] **PC Member**: International Conference on Mining Software Repositories (MSR 2026) (Mining Challenge Track)

[5] **Junior PC Member**: International Conference on Mining Software Repositories (MSR 2026)

[2025]

[4] **Shadow PC Member**: International Conference on Software Engineering (ICSE 2025)

[3] **Junior PC Member**: International Conference on Mining Software Repositories (MSR 2025)

[2] **PC Member**: Software Analytics Research (SOAR) Symposium 2025

[2024]

[1] **Junior PC Member**: International Conference on Mining Software Repositories (MSR 2024)

Journal Reviewer (6)

[6] **ACM Transactions on Software Engineering and Methodology (TOSEM)**
Reviewer since 2023
Reviewed 05 manuscripts

[5] **Empirical Software Engineering (EMSE)**
Reviewer since 2023
Reviewed 04 manuscripts

[4] **Journal of Systems and Software (JSS)**
Reviewer since 2024
Reviewed 03 manuscripts

[3] **IEEE Transactions on Software Engineering (TSE)**
Reviewer since 2025
Reviewed 02 manuscripts

[2] **Journal of Software: Evolution and Process (JSEP)**
Reviewer since 2025
Reviewed 02 manuscripts

[1] **Information and Software Technology (IST)**
Reviewer since 2025
Reviewed 01 manuscript

Sub-reviewer (2018 – 2026)

- International Conference on Software Engineering (ICSE)
- International Conference on Automated Software Engineering (ASE)
- International Conference on Software Maintenance and Evolution (ICSME)
- International Conference on Software Analysis, Evolution and Reengineering (SANER)
- International Working Conference on Source Code Analysis and Manipulation (SCAM)
- International Conference on Program Comprehension (ICPC)
- Innovations in Software Engineering Conference (ISEC)
- International Conference on Collaborative Advances in Software and Computing (CASCON)

LEADERSHIP, SERVICE, AND COMMUNITY ENGAGEMENT (31)

[31] **[2026–Current] Member**, Undergraduate Research Ethics Committee (UREC), Lakehead University, Canada. One of three faculty members reviewing undergraduate research proposals involving human participants for compliance with ethical standards, participant welfare, and responsible research practice.

[30] **[2025] Mentor**, AI4ALL Summer 2025 Program. Mentored two groups (five students) in a national U.S. nonprofit initiative that broadens participation in responsible AI among underrepresented communities.

[29] **[2025] Distinguished Judge**, ACM Student Research Poster Competition, ICSE 2025, Ottawa, Canada.

[28] **[2025] Distinguished Reviewer**, 9th Symposium on Innovations in Computer Science and Applied Computing (ICSAC 2025), University of Saskatchewan, Canada.

[27] **[2024] Distinguished Judge**, USSU Undergraduate Symposium 2024, University of Saskatchewan, Canada.

[26] **[2023] Distinguished Judge**, SOAR Symposium 2023, University of Saskatchewan, Canada.

[25] **[2022] Lead Organizer**, 6th Symposium on Innovations in Computer Science and Applied Computing (ICSAC, Research Fest 2022), Computer Science Graduate Council (CSGC), University of Saskatchewan, Canada.

[24] **[2021–2024] Executive Member (elected)**, Khulna University Alumni Association, Canada.

[23] **[2021–2022] President**, Computer Science Graduate Council (CSGC), Department of Computer Science, University of Saskatchewan, Canada.

[22] **[2021] Program Convener**, Pahela Baishakh and Eid Adda, Khulna University Alumni Association, Canada.

[21] **[2021] Member, Technical Support Team**, SSPP Fund Raising Banquet, Saskatoon, Canada.

[20] **[2020–2024] Social Chair**, Software Research Lab, Department of Computer Science, University of Saskatchewan, Canada.

[19] **[2020–2021] GSA Representative**, Computer Science Graduate Council (CSGC), Department of Computer Science, University of Saskatchewan, Canada.

[18] **[2020–2021] Webmaster**, IEEE Canada North Saskatchewan Section.

[17] **[2019–2020] Vice President Social**, Computer Science Graduate Council (CSGC), Department of Computer Science, University of Saskatchewan, Canada.

[16] **[2015–Current] Chief Advisor**, Tech For Mankind Bangladesh.

[15] **[2015–2018] Coordinator**, Google Developers Group (GDG) Bangla, Khulna University Chapter, Bangladesh.

[14] **[2014–2018] Mentor**, Bangladesh Association of Software and Information Services (BASIS), Khulna University Chapter, Bangladesh.

[13] **[2014–2015] Executive Member (elected)**, Khulna University Teachers' Association, Khulna University, Bangladesh.

[12] **[2013–2014] Web Chair**, 16th International Conference on Computer and Information Technology (ICCIT 2013), Bangladesh.

[11] **[2013–2014] Web Admin**, Khulna University Website, Khulna University, Bangladesh.

[10] **[2012–2018] Founder and Director**, Suresh Sreety Shikkha Niketon, an honorary free school for underprivileged children, Batiaghata, Khulna, Bangladesh.

[9] **[2012–2018] Programming Contest Coordinator and Coach**, Department of Computer Science and Engineering, Khulna University, Bangladesh.

[8] **[2012–2018] Advisor**, Club for Updated Search on Computer (CLUSTER), Khulna University, Bangladesh.

[7] **[2012–2018] Advisor**, Seminar Library, Department of Computer Science and Engineering, Khulna University, Bangladesh.

[6] **[2012–2018] Principal Coordinator**, Science and Information Technology Club (SAiTEC), Secondary and Higher Secondary Schools, Batiaghata, Khulna, Bangladesh.

[5] **[2012–2018] Member**, Volunteers of Bangladesh and Poriborton Chai Bangladesh.

[4] **[2012–2016] Assistant Director of Students' Affairs**, Khulna University, Bangladesh.

[3] **[2012–2015] Program Admin**, Microsoft Developer Network (MSDN) Academic Alliance, Department of Computer Science and Engineering, Khulna University, Bangladesh.

[2] **[2012–2015] Sports Coordinator**, Department of Computer Science and Engineering, Khulna University, Bangladesh.

[1] **[2010–2012] Program Coordinator and Volunteer**, Computers Are Free For Everyone (CAFFE), Dhaka, Bangladesh. Taught computer skills to eleven street children within one year, enabling them to earn a living.

TALKS, POSTERS, AND DEMOS (21)

[21] **Saikat Mondal**. 2025, "Can We Trust the AI Pair Programmer? Copilot for API Misuse Detection and Correction", **CASCON 2025**, Toronto, Canada.

[20] **Saikat Mondal**. 2025, "Does Editing Improve Answer Quality on Stack Overflow? A Data-Driven Investigation", ICSAC 2025, Saskatoon, Canada.

[19] **Saikat Mondal**. 2025, "Transforming Machine Intelligence: From ML to GenAI", DevFest 2025, Google Developer Group (GDG), Saskatoon, Canada. (**Invited Talk**)

[18] **Saikat Mondal**. 2025, "GENCNIPPET: Automated Generation of Code Snippets for Supporting Programming Questions", **MSR 2025**, Ottawa, Canada.

[17] **Saikat Mondal**. 2024, "Enhancing User Interaction in ChatGPT: Characterizing and Consolidating Multiple Prompts for Issue Resolution", **MSR 2024**, Lisbon, Portugal.

[16] **Saikat Mondal**. 2024, "Investigating the Utility of ChatGPT in the Issue Tracking System: An Exploratory Study", **MSR 2024**, Lisbon, Portugal.

[15] **Saikat Mondal**. 2023, "Investigating Technology Usage Span by Analyzing Users' Q&A Traces in Stack Overflow", **APSEC 2023**, Seoul, South Korea.

[14] **Saikat Mondal**. 2023, "Automatic Prediction of Rejected Edits in Stack Overflow", **ICSE 2023** (Journal First Track), Melbourne, Australia.

[13] **Saikat Mondal**. 2023, "Do Subjectivity and Objectivity Always Agree? A Case Study with Stack Overflow Questions", **MSR 2023**, Melbourne, Australia.

[12] **Saikat Mondal**. 2022, "Reproducibility Challenges and Their Impacts on Technical Q&A Websites: The Practitioners' Perspectives", **ISEC 2022**, DA-IICT Gandhinagar, India.

[11] **Saikat Mondal**. 2021, "Rollback Edit Inconsistencies in Developer Forum", **MSR 2021**, Madrid, Spain.

[10] **Saikat Mondal**. 2021, "The Reproducibility of Programming-Related Issues in Stack Overflow Questions", Computer Science and Bioinformatics Symposium, University of Saskatchewan, Canada. (**Poster**, People's Choice Award)

[9] **Saikat Mondal**. 2021, "Early Detection and Guidelines to Improve Unanswered Questions on Stack Overflow", **ISEC 2021**, Bhubaneswar, India.

[8] **Saikat Mondal**. 2020, "Automatic Identification of Rollback Edit with Reasons in Stack Overflow Q&A Site", **ICSME 2020**, Adelaide, Australia.

[7] **Saikat Mondal**. 2019, "Can Issues Reported at Stack Overflow Questions be Reproduced? An Exploratory Study", **MSR 2019**, Montreal, Canada.

[6] **Saikat Mondal**. 2016, "Smartphone-Based Ischemic Heart Disease (Heart Attack) Risk Prediction using Clinical Data and Data Mining Approaches, a Prototype Design", ICCIT 2016, Dhaka, Bangladesh.

[5] **Saikat Mondal**. 2016, "An Improved Color Image Steganography Technique in Spatial Domain", ICECE 2016, Dhaka, Bangladesh.

[4] **Saikat Mondal**. 2015, "Innovations for Service", Divisional ICT Fair, Ministry of ICT, Bangladesh. (**Invited Talk**)

[3] **Saikat Mondal**. 2014, "A Robust RGB Channel Based Image Steganography Technique using a Secret Key", ICCIT 2013, Khulna, Bangladesh.

[2] **Saikat Mondal**. 2014, "PHP for Software Development", Google Developers Group (GDG) Bangla, Khulna University Chapter, Bangladesh. (**Invited Talk**)

[1] **Saikat Mondal**. 2010, "A New Approach of Extendable Multicast Routing Protocol in MANET", ICCIT 2010, Dhaka, Bangladesh.

PROFESSIONAL PROJECTS (5)

[5] **Home Entertainment WLAN SoC Host Software Development**, Connectivity Division, Samsung R&D Institute Bangladesh.

[4] **Spyder Wi-Fi Display**, Connectivity Division, Samsung R&D Institute Bangladesh.

[3] **DSP Software Development for Camera and Channel (Phases I and II)**, Reconfigurable Processor Division, Samsung R&D Institute Bangladesh.

[2] **IPC Mailbox System for VxWorks**, Connectivity Division, Samsung R&D Institute Bangladesh.

[1] **Website Design and Development**, Khulna University and ICCIT 2013, Bangladesh.

SOFTWARE TOOLS AND PROTOTYPES (8)

[8] **OpenART**
Overview Manuscript-driven support that promotes the executability of software engineering research artifacts.
Publications JAWs 2026

[7] **SciFlowgent**
Overview Agentic AI support for developing scientific workflows.
Publications JAWs 2026

[6] **SafeSnippet**
Overview A browser-integrated tool that uses GPT-4.1 with retrieval-augmented generation to detect and fix security vulnerabilities (CWEs) in noisy, parsable and non-parsable Python code from Stack Overflow.
Publications In preparation

[5] **AUTOCOMBAT**
Overview Human-aligned enhancement of programming answers with LLMs guided by user feedback.
Publications TOSEM (under revision)

[4] **GENCNIPPET**
Overview An LLM-powered tool that generates relevant code examples for Stack Overflow questions during submission.
Publications MSR 2025 (Registered Reports Track)

[3] **AUTOGENICS**
Overview Automated generation of context-aware inline comments for code snippets on programming Q&A sites using LLMs.
Publications SCAM 2024

[2] **iEdit**
Overview An online tool that detects inconsistent edits automatically and guides Stack Overflow editors with suggestions and rejection ratios.
Publications MSR 2021 and Automated Software Engineering Journal 2026

[1] **EditEx**
Overview A tool that identifies edits likely to be rejected, with potential reasons, and assists users during real-time editing on Stack Overflow.
Publications ICSME 2020 (Registered Reports Track) and EMSE 2022

PROFESSIONAL TRAINING (8)

[8] **Mental Health First Aid**, University of Saskatchewan (supported by the Mental Health Commission of Canada), Canada.

[7] **Quantum Computing** (one-month course), Centre for Development of Advanced Computing (C-DAC) in collaboration with the Indian Institute of Technology Roorkee, Ministry of Electronics and Information Technology, Government of India.

[6] **Teaching Pedagogy (Induction, Modules I and II)**, Center of Excellence in Teaching and Learning and Institutional Quality Assurance Cell, Khulna University, Bangladesh.

[5] **Teaching Methodology**, Research Cell, Khulna University, Bangladesh.

[4] **Research Methodology**, Higher Education Quality Enhancement Project, Economics Discipline, Khulna University, Bangladesh.

[3] **Child Rights**, Save the Children, Bangladesh.

[2] **Mobile Application Development**, Ministry of Information and Communication Technology (MoICT), Bangladesh.

[1] **Process Engineering**, Samsung R&D Institute Bangladesh.

RESEARCH TOOLS AND TECHNOLOGY EXPERIENCE

- **Large Language Models and Generative AI:** GPT-4/4o/5, LLaMA 2/3, DeepSeek-Coder, CodeLlama, StarCoder, Mixtral, prompt engineering, chain-of-thought design, structured prompting, safety alignment, controllable generation, and multi-model evaluation.
- **Retrieval-Augmented Generation (RAG):** Embedding models, vector databases (FAISS, Chroma, Pinecone), document chunking, hybrid retrieval, grounding strategies, context optimization, retrieval evaluation, and RAG pipeline development.
- **Fine-Tuning and Model Adaptation:** LoRA, QLoRA, PEFT, instruction tuning, domain adaptation, supervised fine-tuning, dataset curation, prompt–response pair design, model evaluation (BLEU, ROUGE, CodeBLEU), and safety and robustness tuning.
- **Agentic AI and Autonomous Code Tools:** Multi-agent frameworks, task decomposition, generalist and specialist agent design, tool-augmented reasoning, program repair agents, diagnostic agents (correctness, security, performance), survival scoring, tool calling, and agent orchestration for code intelligence.
- **Machine Learning and Data Mining:** Scikit-learn, WEKA, R, MATLAB, Decision Trees (CART), Random Forest, XGBoost, ANN, KNN, SVM, Logistic Regression, Naive Bayes, Linear Regression, resampling, SMOTE, Bagging, Boosting, Ensemble Learning, SHAP, and LIME.
- **Natural Language Processing:** Stanford CoreNLP, Mallet, ROUGE, POS tagging, sentiment analysis, SentiStrength-SE, SEntiMoji, term weighting, text summarization, discourse analysis, and semantic similarity analysis.
- **Source Code Analysis:** AST analysis, syntactic similarity, readability, understandability, and complexity analysis.
- **Code Search and Information Retrieval:** Lucene.
- **Software Development and Maintenance:** Eclipse, IntelliJ, PyCharm, NetBeans, Visual Studio, Code::Blocks, Jupyter Notebook, JUnit, Maven, JavaParser, Jsoup, PMD, FindBugs, Roslyn, and Esprima.
- **Software Version Control:** Git and GitHub.
- **Statistics and Data Modeling:** Probability distributions, random sampling, confidence intervals, central tendency, data centrality, and statistical significance tests.
- **Data Visualization:** Gephi and OGDF.
- **Programming Languages:** Java, Python, and C/C++.
- **Reporting and Prototyping:** LaTeX, Adobe Photoshop, MS Office, and Pencil.
- **Research Collaboration:** Overleaf, Slack, Microsoft Teams, and WhatsApp.

PROFESSIONAL REFERENCES

(1) Dr. Chanchal K. Roy

Professor, University of Saskatchewan, Canada
Email: chanchal.roy@usask.ca
URL: https://www.cs.usask.ca/faculty/croy

(2) Dr. Gias Uddin

Associate Professor, York University, Canada
Email: guddin@yorku.ca
URL: https://lassonde.yorku.ca/users/guddin

(3) Dr. Masud Rahman

Associate Professor, Dalhousie University, Canada
Email: masud.rahman@dal.ca
URL: https://web.cs.dal.ca/~masud

Version: *October 2026*